Life Sciences and
Facility Management

Institute of
Computational Life Sciences

Feature matching, image registration, stereo vision

Image and Signal Processing

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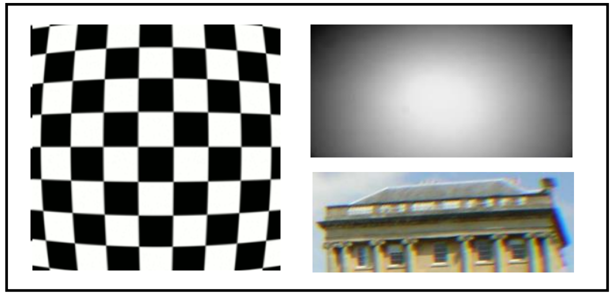
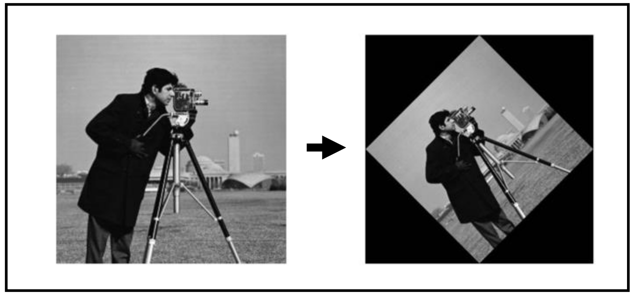
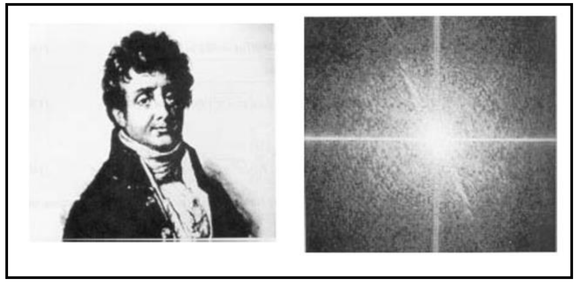
Recap of last weeks' topics

- Image quality attributes
- Image transformations
- Fourier transform / spectral filtering
- Spatial filtering
- Edge detection and features







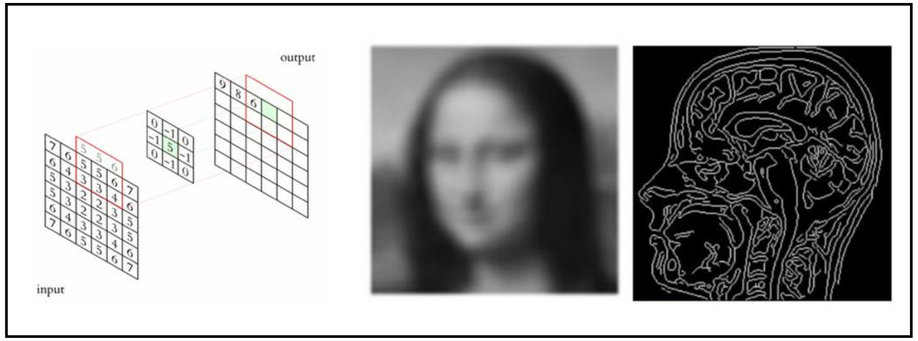
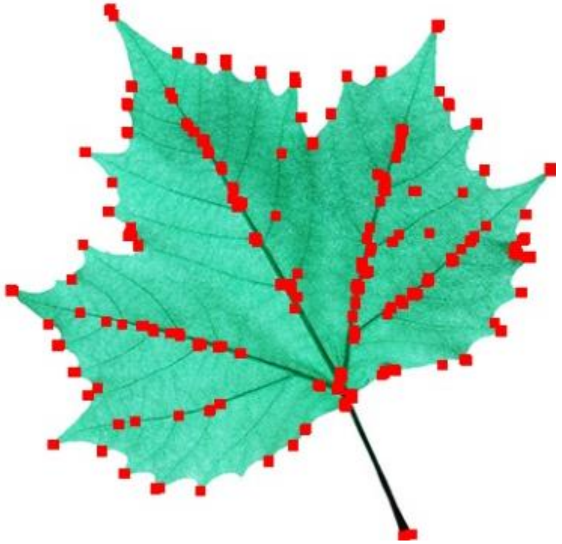


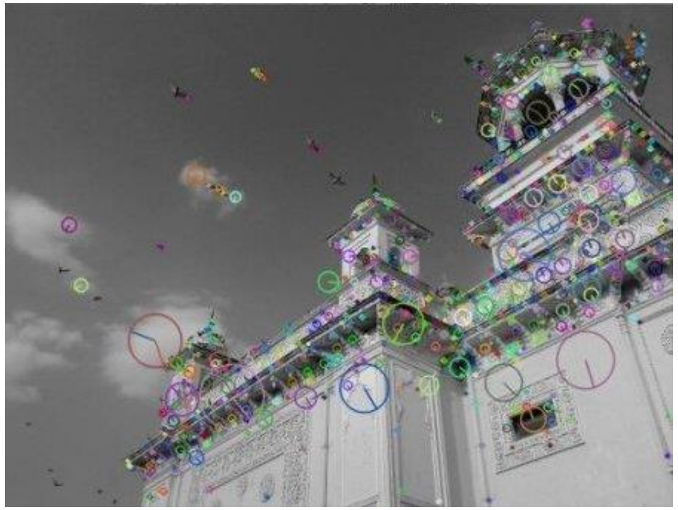
Illustration: Link

Feature matching

Recap: Detection of key points

- Key points (or points of interest, salient points) are points in the image with distinctive characteristics
- Examples: Harris corners, SIFT or SURF features
- Ideally, key points are robust to changes in the image, such as rotation, scaling, translation, noise, illumination changes, etc.





Examples for key points / point features:
Harris corners (top), SIFT features bottom

Recap: Detection of key points

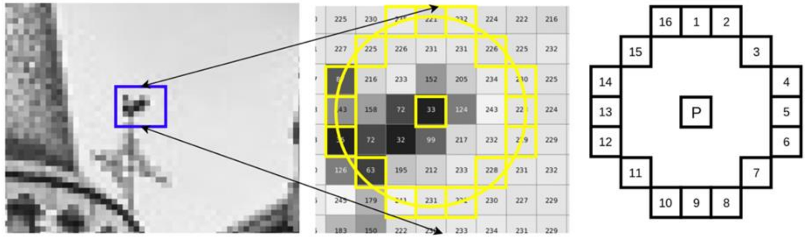
- Key points (or points of interest, salient points) are points in the image with distinctive characteristics
- Examples: Harris corners, SIFT or SURF features
- Ideally, key points are robust to changes in the image, such as rotation, scaling, translation, noise, illumination changes, etc.
- Key points are used in various computer vision tasks, such as image registration, tracking, stereo vision, and object recognition.

Find the differences... Which key points (Shi-Tomasi features) are stable?

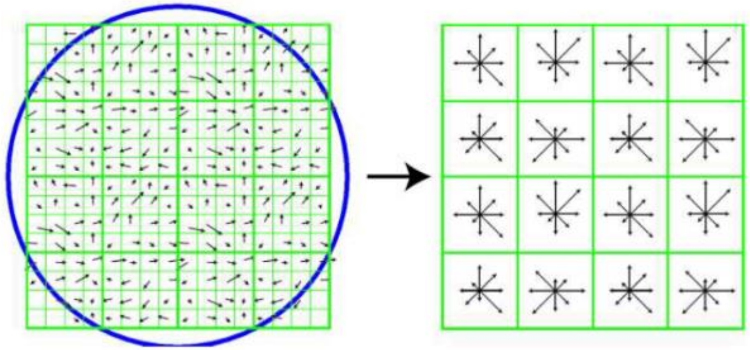
Feature descriptor

- Some key point detection methods come with a **descriptor** that describe the image in the proximity of the feature location
- Concretely: A descriptor is a set of numbers that describes the region around a key point (like appearance, shape, or texture)
- **Idea**: Use descriptors to compare key points with respect to similarity.
- Descriptors usually come with their own distance metric. Often, this is simply the Euclidean norm of the difference between two descriptors:

$$\left\|\vec{d}_a - \vec{d}_b\right\| = \sqrt{(d_{a,0} - d_{b,0})^2 + (d_{a,1} - d_{b,1})^2 + \ldots}$$



FAST



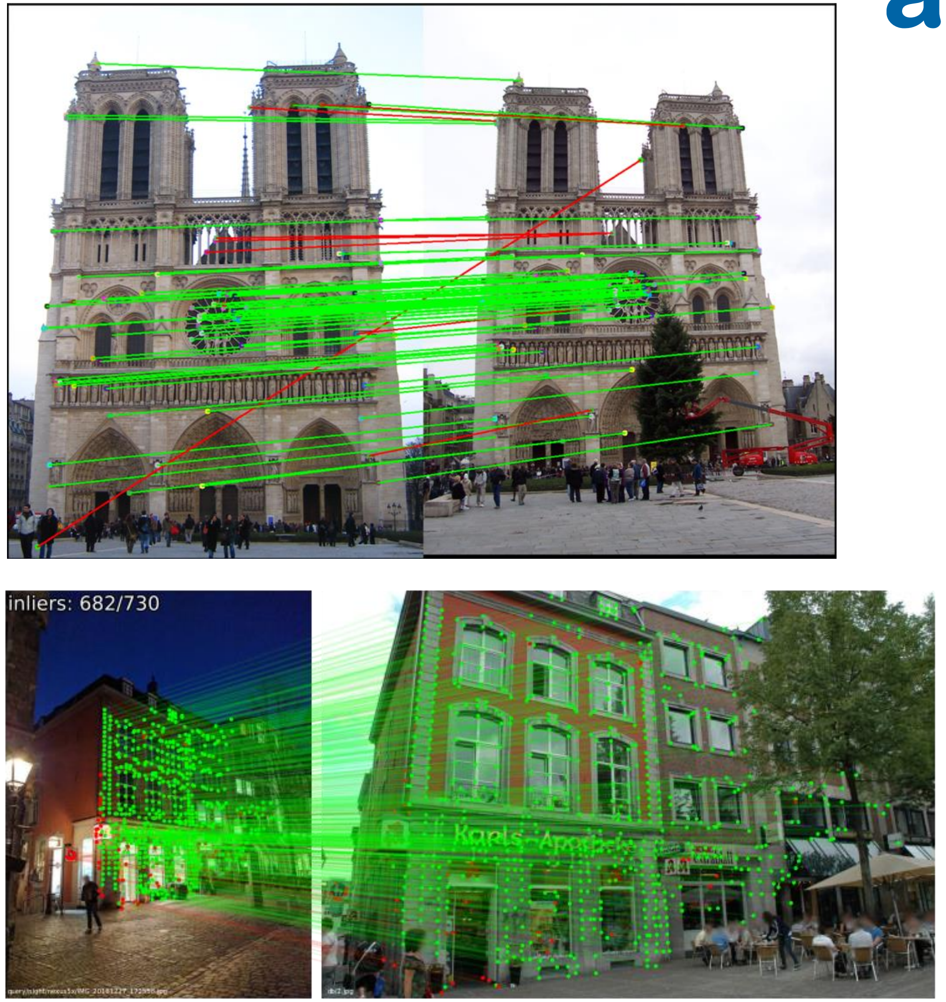
SIFT

Illustrations of different descriptors. Top: FAST feature descriptor. Bottom: SIFT feature descriptor. Details are omitted here

Feature matching

- Feature matching is the process of finding corresponding points in two images.
- Steps:
  1. Detect key points in both images
  2. Compute the descriptors for the key points
  3. Match the descriptors

- Wrong matches (often called outliers) can occur due to:
  - Occlusions or perspective changes
  - Illumination changes
  - Noise (or other distortions)

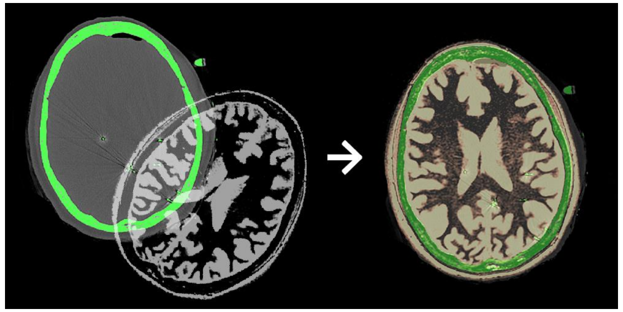


Examples for matching features in two images. Matches are visualized using a green connecting line. Red dots (and lines) identify non- or mis-matching features. Both examples rely on SIFT features.

Image registration

Image registration

- Image registration: Use matching feature pairs to estimate the transformation between two images.
- Transformation: Mathematical rule (function) with which we can project a pixel p of image I_1 into I_2
- Subtypes of image registration:
  - Rigid registration, using
    - similarity transform: Translation, rotation, and scaling
    - affine transform : Translation, rotation, scaling, and shearing
    - projective transform : Perspective transformations
  - Non-rigid registration:
    - Makes use of local deformations
    - Transformation fields



Examples for **rigid** / affine transforms

Image registration

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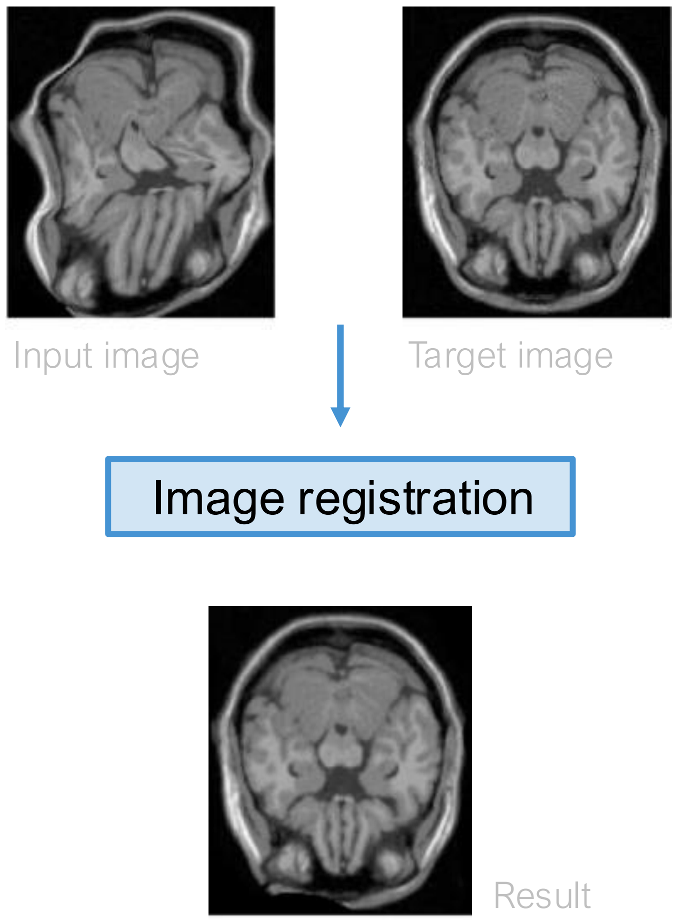
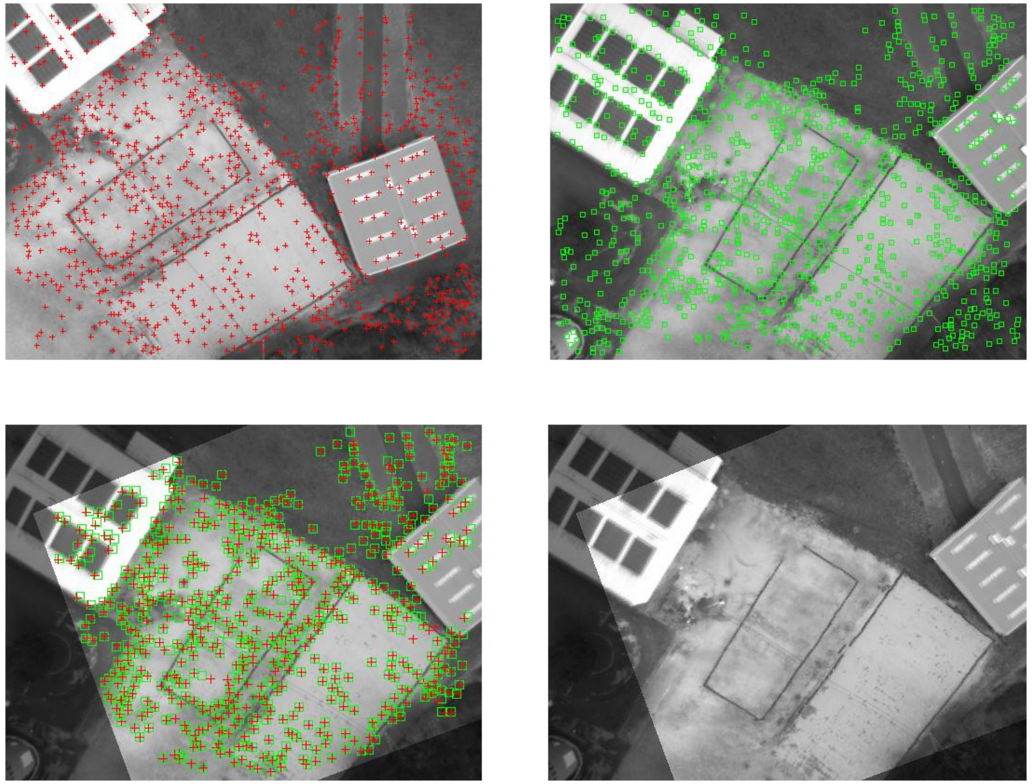


Example for **non-rigid** transform

Image registration

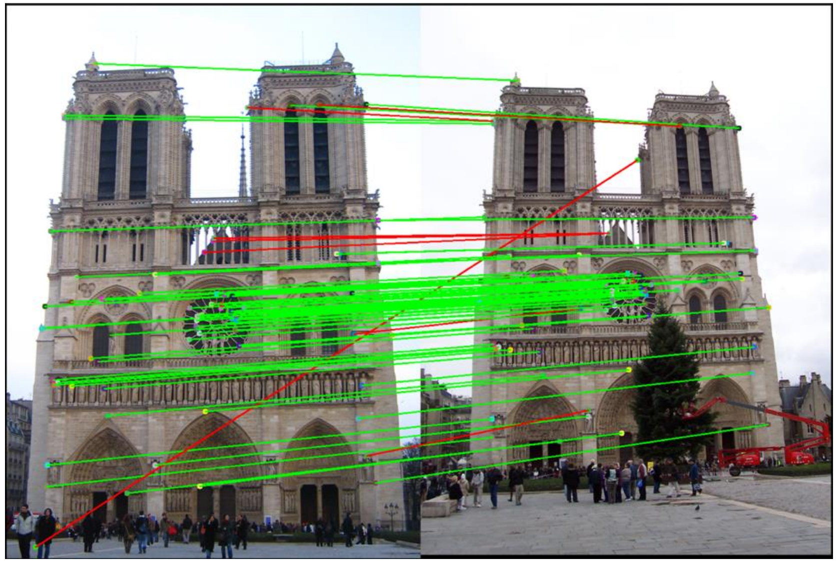
- Steps:
  1. Feature matching
     - Detect key points in both images
     - Compute the descriptors
     - Match the descriptors
  2. Estimate the transformation $T : p \rightarrow q$
     - Often involves solving an optimization problem
     - Variables: Parameters θ of the transformation
     - Input data: Matched pairs of key points: (p_i, q_i)
     - Cost function: mean reprojection error, i.e. distance between q_i and the transformed points p_i : $\mathring{q_i} = T(p_i)$
     - Minimization of cost function yields optimal θ
  3. Apply the transformation to the image



Top: Left image containing key points p_i, and right image containing features q_i. Bottom: matched and projected features after estimation of the (affine) transformation T

Image registration: RANSAC

- How to detect outliers?
- Popular method: RANSAC
  - RANSAC: Random sample consensus
  - Iterative method to estimate the parameters of a mathematical model.
- Steps:
  1. Select a subset of data points randomly
  2. Fit the model to the selected data points
  3. Compute the error of the model on all data points
  4. Count the number of inliers (= points that are consistent with the model)
  5. Repeat steps 1-4 N times, or until convergence
- Extension: PROSAC (Progressive sample consensus). ... ☺



Note the non-matching pairs of points. They have a negative effect on the estimated transformation T

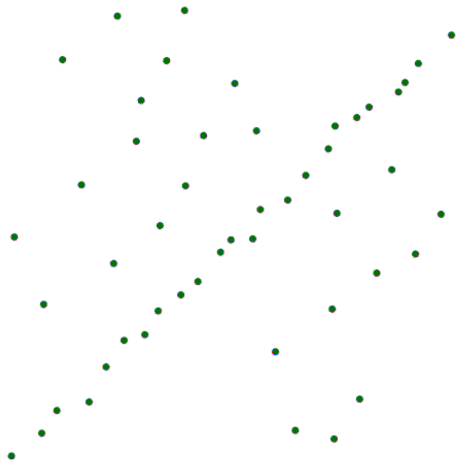
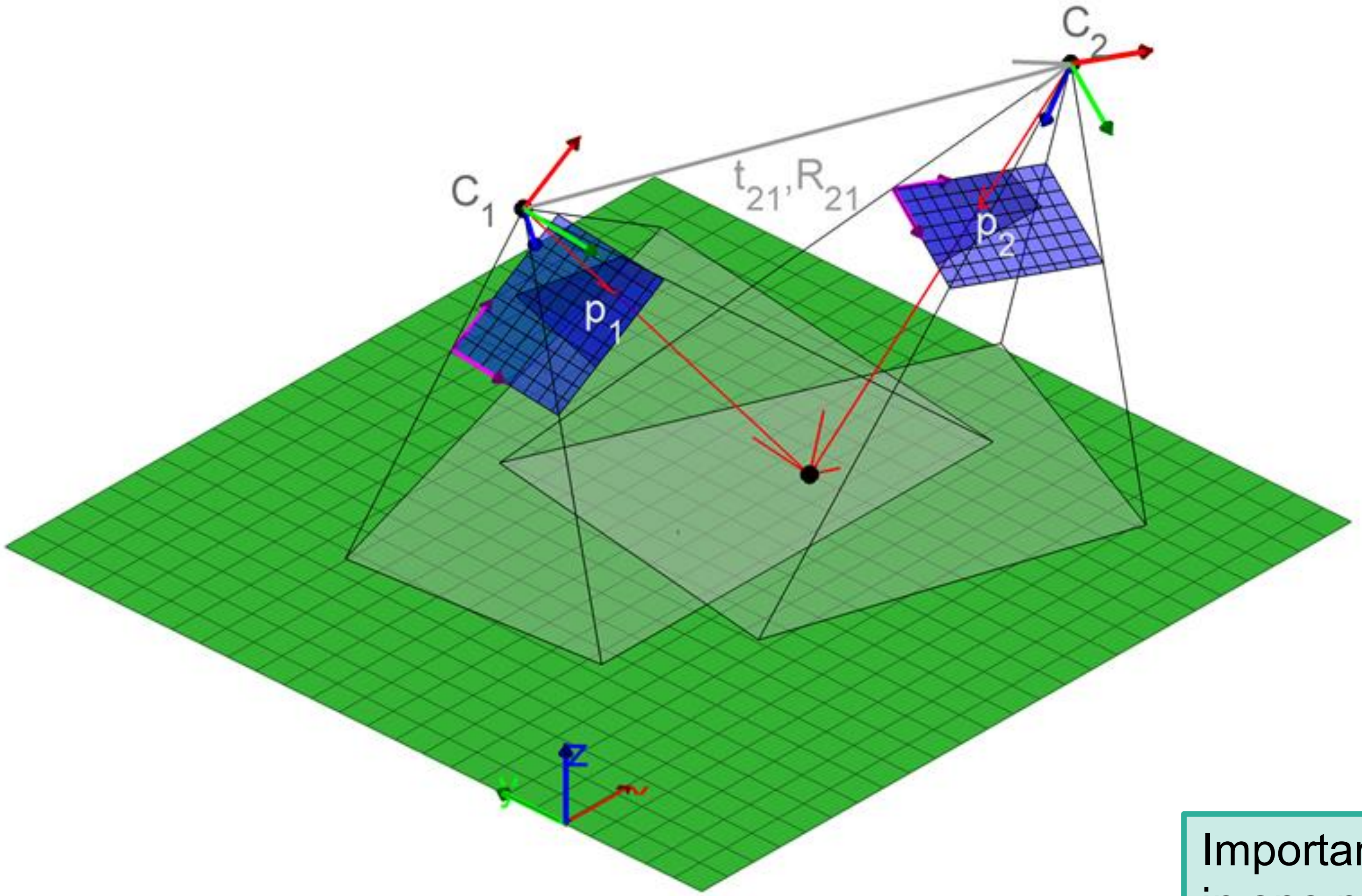


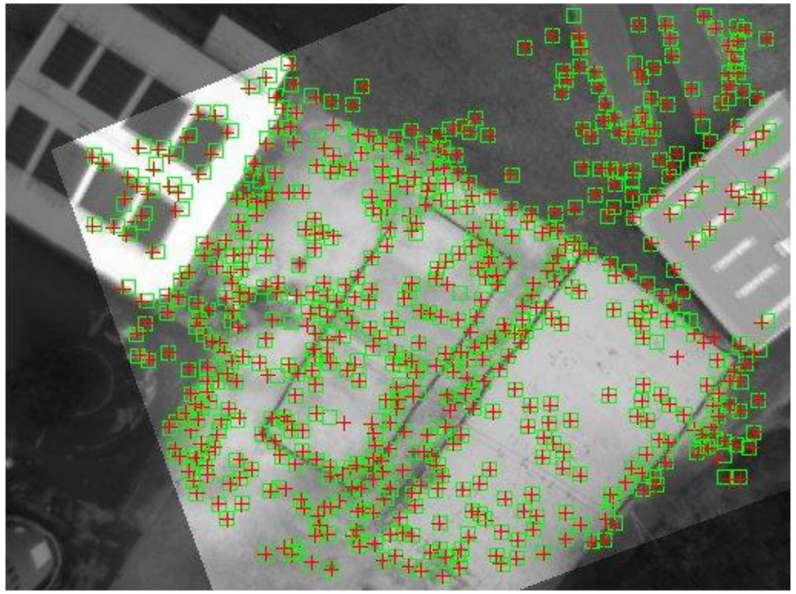
Illustration of the RANSAC algorithm for a line model: two points are sampled randomly, and we track which of the models finds best support in the dataset. Link

Applications

...for feature matching

Application: Aerial image registration

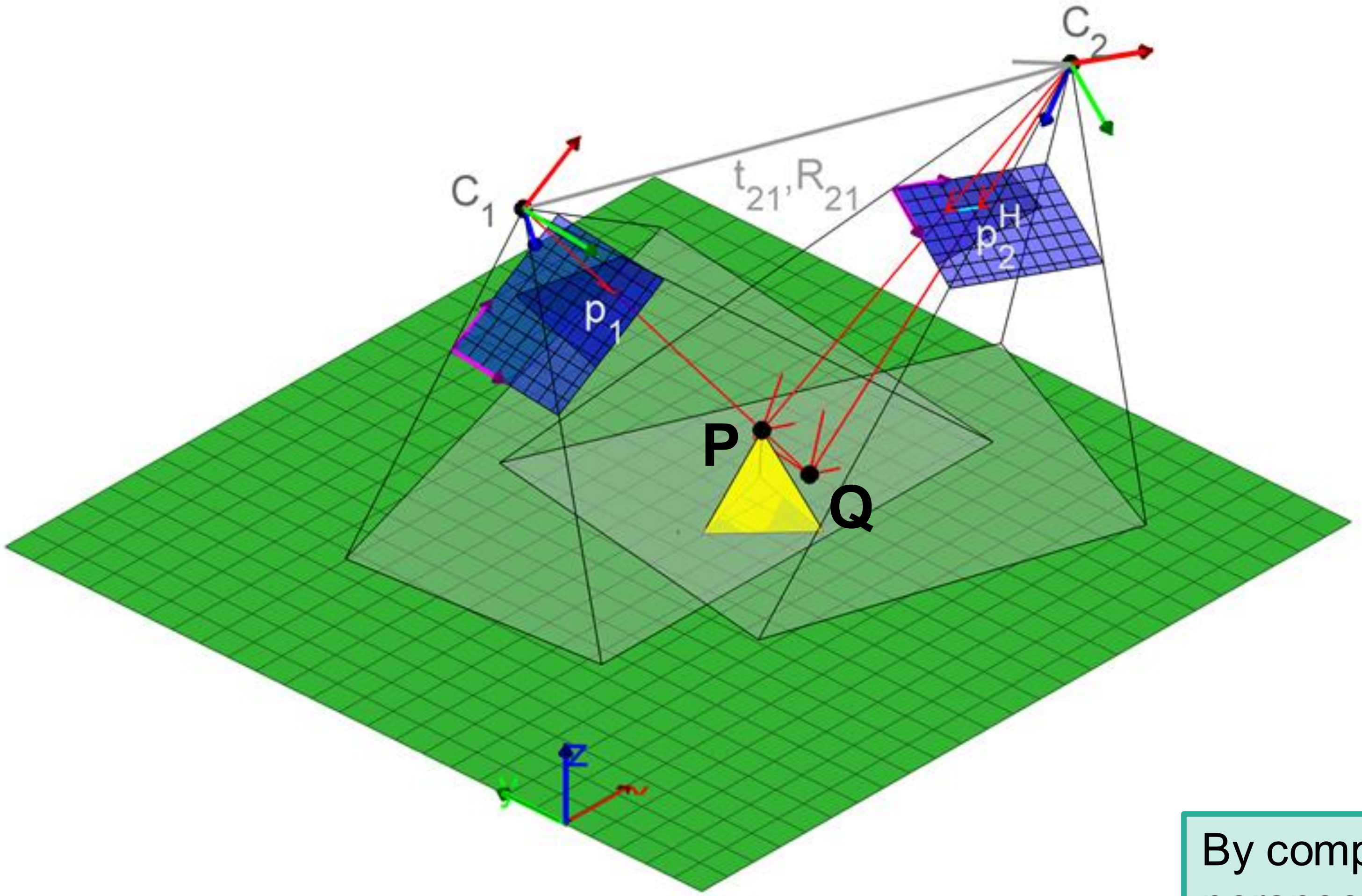




Left: Illustration of two cameras acquiring aerial images from different viewpoints. Top: Result after image registration.

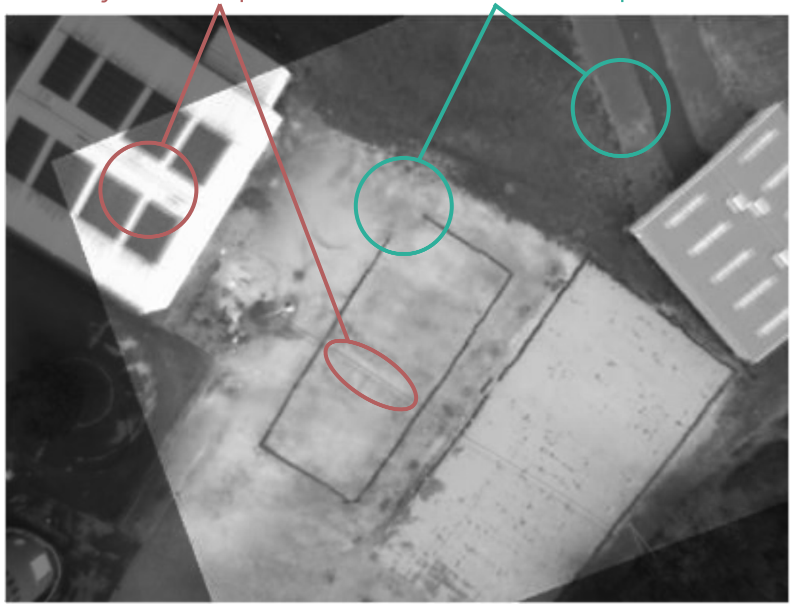
Important assumption: All points lie approximately in one plane. This assumption is more accurate the further away the camera is from the ground.

Application: Stereo vision



Out-of-plane structures appear displaced in the overlay due to parallax.

In-plane structures remain fixed in position.

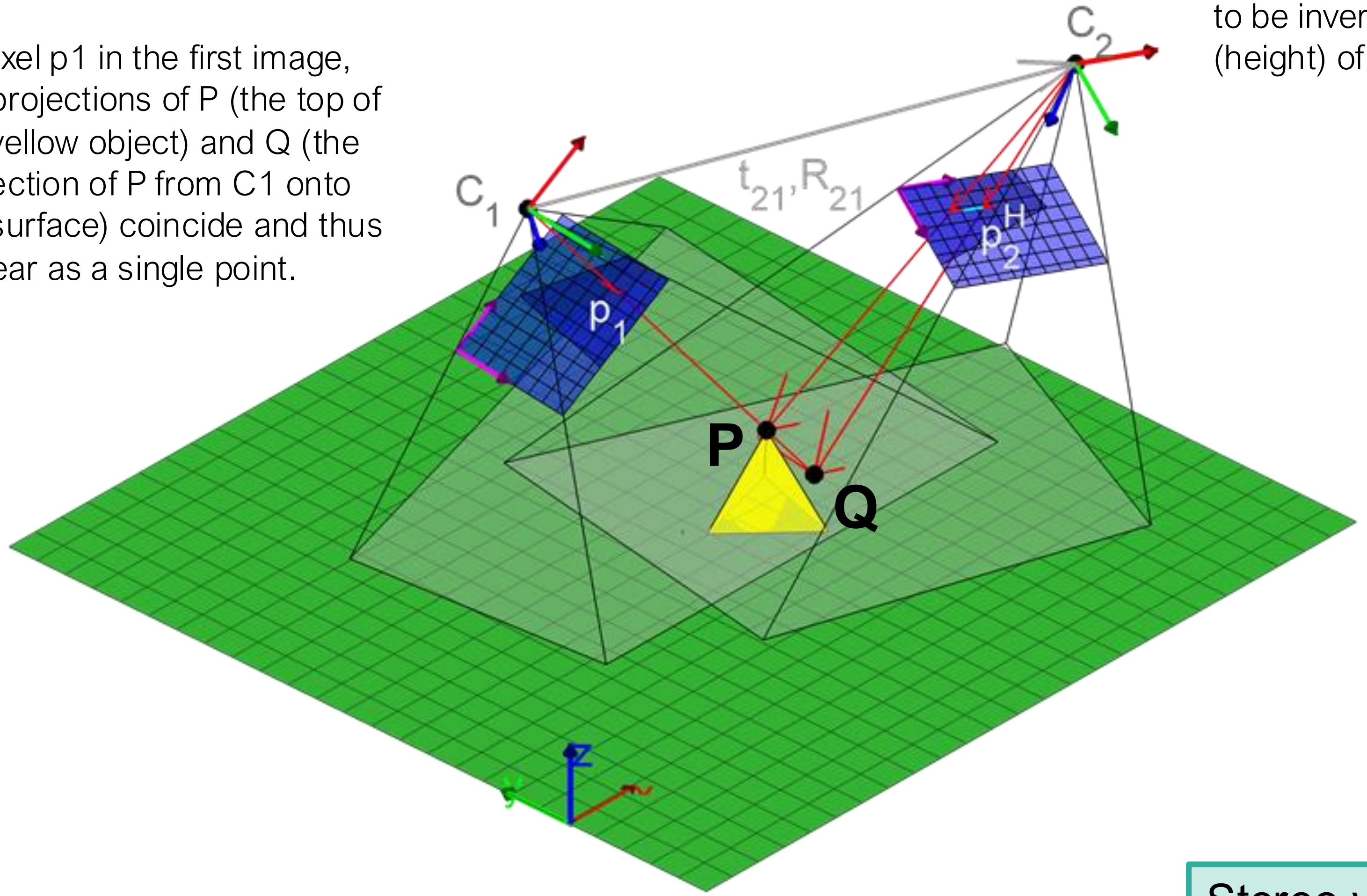


Top: The overlay of two aligned images shows that most pixels in the projected image align well with the target image; however, elevated structures (e.g., trees, roofs) do not.

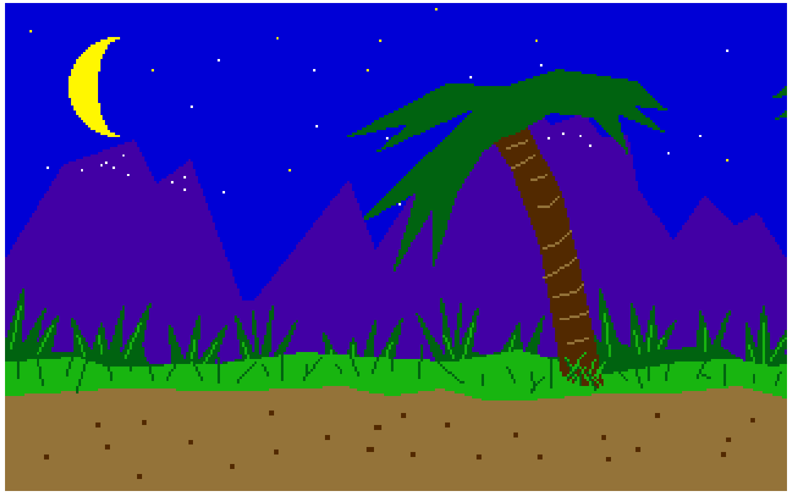
By comparing two images taken from different perspectives, we can infer information about the 3D extent of objects in the scene.

Application: Stereo vision

At pixel p1 in the first image, the projections of P (the top of the yellow object) and Q (the projection of P from C1 onto the surface) coincide and thus appear as a single point.



In image 2, points P and Q appear at different positions. This positional difference is referred to as **disparity**. The disparity can be shown to be inversely proportional to the depth (height) of the yellow object.

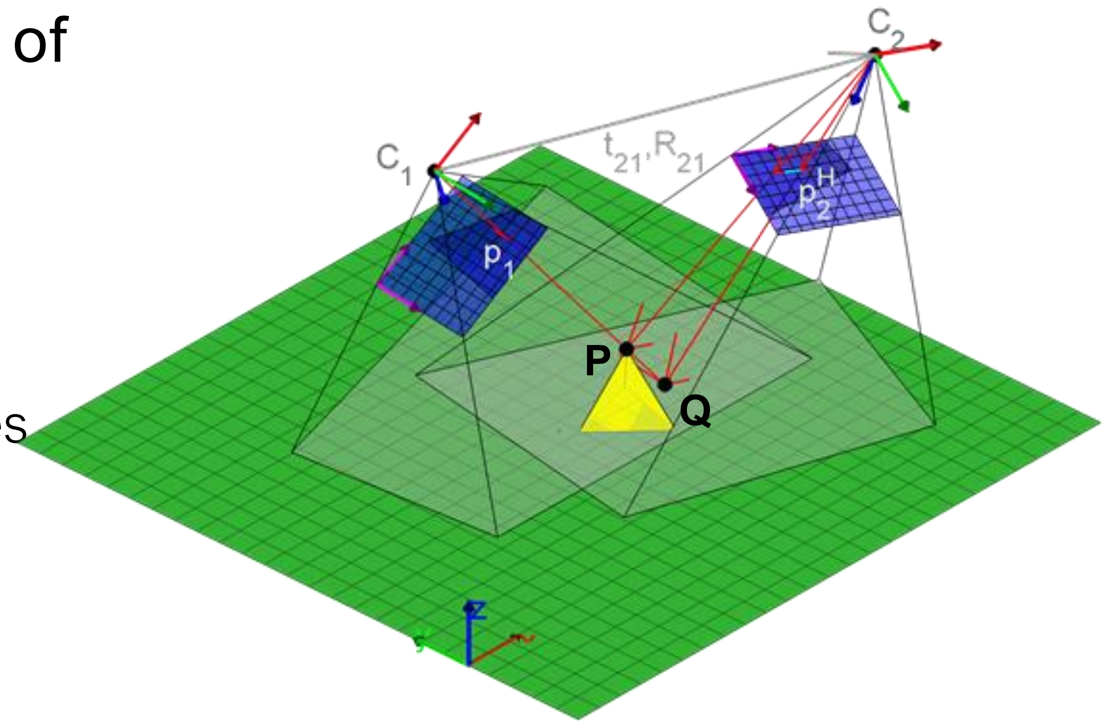


Top: Animation of the parallax effect (link). As the viewpoint changes, distant objects appear to move more slowly than objects closer to the camera. This effect is exploited in stereo vision to estimate object depth.

Stereo vision refers to the process of extracting 3D information from two or more images.

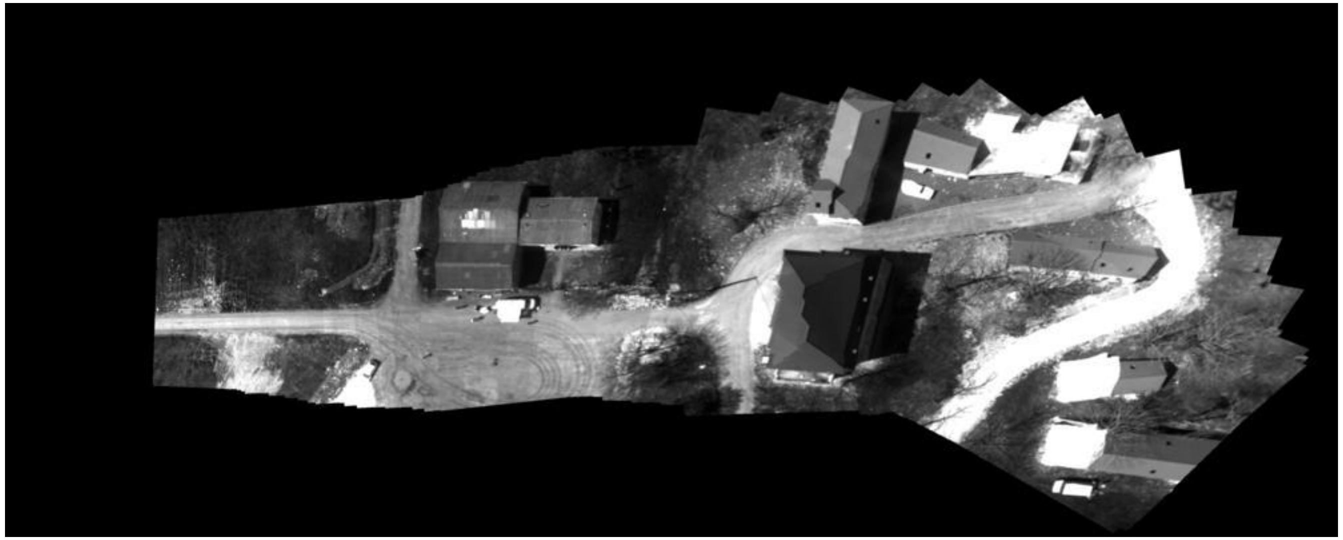
Application: Stereo vision

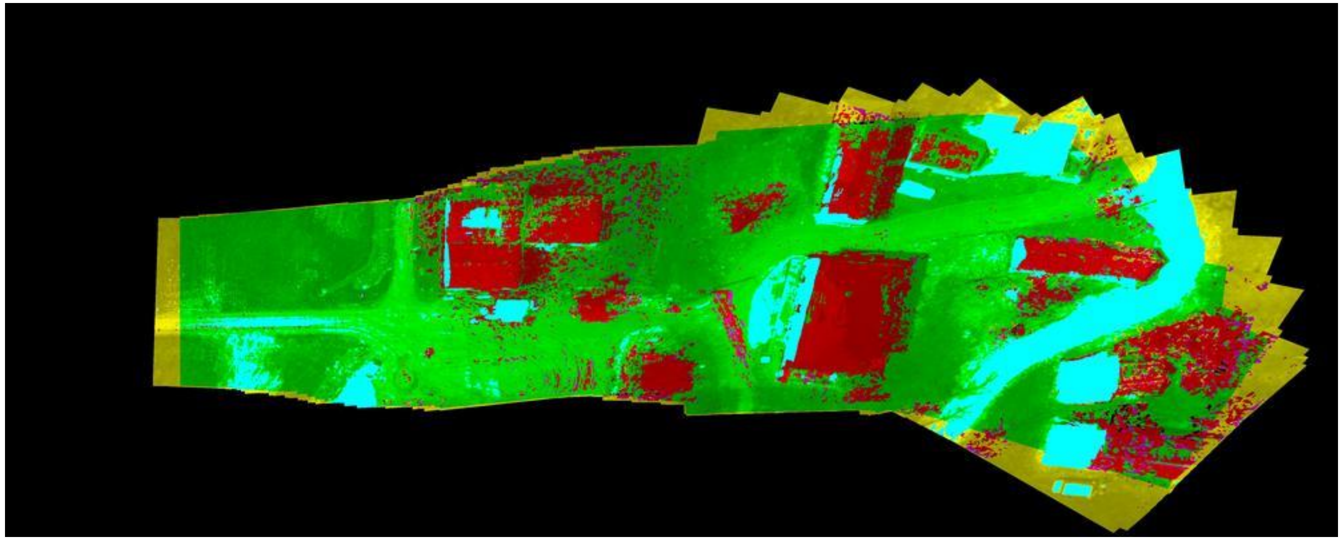
- **Goal** (of stereo vision): Compute the depth of objects in a scene from two (or more) images.
- **Depth map**: An image that encodes the depth of objects in a scene for every pixel.
- Typical steps:
  - Align the images, for example by matching features.
  - Estimate the disparity (displacement) for each pixel.
  - Compute the depth map from the estimated disparities.



Application: Traversability mapping

- Stereo vision can be used to assess ground traversability for a ground vehicle.
- Input: Sequence of aerial images
- Steps:
  - Detect and match features
  - Align the images
  - Compute the depth map (estimate depth at each pixel).
  - Threshold the depth map to identify obstacles and free space.
- Output:
  - A map indicating passable areas





Application: Image stitching / Panography

- Image stitching is the process of combining multiple images into a single composite image.

- Panography is a form of image stitching that produces a wide-angle panoramic image from multiple images.

Half of the images aligned

All images aligned using bundle adjustment

After photometric alignment, seam selection and blending

a) Images are geometrically aligned through image stitching.
b) Bundle adjustment then optimizes the alignment globally, rather than sequentially on an image-by-image basis.
c) Photometric alignment compensates for brightness and exposure differences between images, while blending further reduces visible seams to produce a seamless result.

Source: https://www.doi.org/10.1007/978-0-387-31439-6